GCWUS

Web Design & Development

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Lab Manual

- Make calculator using Html, CSS and JavaScript.

CALCULATOR

Q: *Make calculator using Html, CSS and JavaScript.*

HTML CODE:

assignment2.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Calculator</title>
    <link rel="stylesheet" href="assignment2.css">
</head>
<body>
    <div class="calculator">
  <!-- The screen where results appear -->
        <input type="text" id="display" disabled>
  <!-- The container for all buttons -->
        <div class="buttons">
   <!-- Buttons use 'onclick' to run JS functions -->
            <button class="clear" onclick="clearDisplay()">C</button>
            <button class="operator" onclick="appendToDisplay('/')">/</button>
            <button class="operator" onclick="appendToDisplay('*')">*</button>
            <button class="operator" onclick="appendToDisplay('-')">-</button>
            <button onclick="appendToDisplay('7')">7</button>
            <button onclick="appendToDisplay('8')">8</button>
            <button onclick="appendToDisplay('9')">9</button>
            <button class="operator" onclick="appendToDisplay('+')">+</button>
            <button onclick="appendToDisplay('4')">4</button>
            <button onclick="appendToDisplay('5')">5</button>
            <button onclick="appendToDisplay('6')">6</button>
            <button class="equals" onclick="calculate()">=</button>
            <button onclick="appendToDisplay('1')">1</button>
            <button onclick="appendToDisplay('2')">2</button>
            <button onclick="appendToDisplay('3')">3</button>
            <button onclick="appendToDisplay('0')">0</button>
            <button onclick="appendToDisplay('.')">.</button>
        </div>  </div>
    <script src="assignment2.js"></script>
</body>
</html>
```

CSS CODE:

assignment2.css > ...

```css
body {
    margin: 0;  place-items: center;
    height: 100px;
    background: #f0f0f0;
    font-family: Arial, sans-serif;
}
.calculator, #display {
    background: #fff;
    border-radius: 10px;
}
.calculator {
    padding: 20px; width: 300px;
}
#display {
    width: 100%; height: 40px;
```

```css
    margin-bottom: 20px;
    padding: 10px; font-size: 24px;
    text-align: right;  border: none;
    box-sizing: border-box;
}
.buttons {
    display: grid;  gap: 10px;
    grid-template-columns: repeat(4, 1fr);
}
button {
    height: 40px;  border: none;
    border-radius: 5px;  background: #e0e0e0;
    cursor: pointer; font-size: 18px;
}
button:hover { background: #d0d0d0; }
.clear { background: #f66; color: #fff; }
.operator { background: #6c0; color: #fff; }
.equals { background: #66f; color: #fff;
 grid-row: span 2; }
```

JAVASCRIPT CODE:

```js
assignment2.js > ...
let display = document.getElementById('display');
// Function to add a number or operator to the screen
function appendToDisplay(value) {
    display.value += value; }
// Function to erase the screen
function clearDisplay() {
    display.value = '';
}
// Function to calculate the final result
function calculate() {
    try {
//  'eval()' reads the string on the display and does the math
        display.value = eval(display.value);
    } catch (error) {
        display.value = 'Error';
// If the user types bad math (like "5*/"), show Error
    } }
```

CODE DETAILS:

⇒ HTML Code Details

Element/Tag	Description/Purpose
`<div class="calculator">`	A container `div` that wraps the entire calculator structure.
`<input type="text" id="display" disabled>`	The calculator's screen/display area, which is disabled from user input.
`<div class="buttons">`	A container for all the calculator buttons.
`<button class="clear">`, `<button class="equals">`	Specific buttons for clearing the display ("C") and calculating the result ("=").
`<button class="operator">`	Buttons for mathematical operations (`/` , `*` , `-` , `+`).
`<button>`	Buttons for numeric digits (0-9) and the decimal point (`.`).
`onclick="..."`	JavaScript event handlers that call functions when a button is clicked.

⇒ CSS Code Details

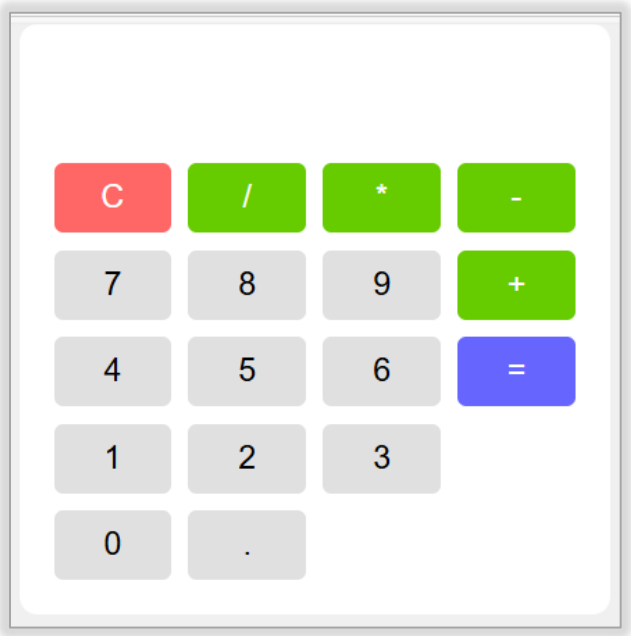
Selector	Property	Description/Purpose
`body`	`margin: 0;`	Removes default browser margins around the body of the page.
	`place-items: center;`	Centers content vertically and horizontally within the body.
	`height: 100vh;`	Sets the body height to 100% of the viewport height.
	`background: #f0f0f0;`	Sets a light gray background color for the page.
	`font-family: Arial, sans-serif;`	Specifies the default font for the entire page.
`.calculator`	`padding: 20px;`	Adds 20 pixels of space inside the calculator container, between the edge and its content.
	`width: 300px;`	Sets the overall width of the calculator container to 300 pixels.
`#display`	`width: 100%;`	Makes the display input field take the full width of its parent container.
	`height: 40px;`	Sets the height of the display field.
	`margin-bottom: 20px;`	Adds 20 pixels of space below the display field.
	`padding: 10px;`	Adds 10 pixels of space inside the display field.
	`font-size: 24px;`	Sets the text size within the display field.
	`box-sizing: border-box;`	Ensures padding and borders are included in the element's total width and height calculation.
`.buttons`	`display: grid;`	Establishes a grid layout for the button container.
	`gap: 10px;`	Sets the space (gutter) between the grid items (buttons) to 10 pixels.
	`grid-template-columns: repeat(4, 1fr);`	Creates four columns of equal fractional width (1fr each) within the grid.
`button`	`height: 40px;`	Sets the height of all buttons.
	`border: none;`	Removes default borders from buttons.
	`cursor: pointer;`	Changes the mouse cursor to a hand icon when hovering over a button.
	`font-size: 18px;`	Sets the text size for buttons.
`button:hover`	`background: #d0d0d0;`	Changes the background color of a button when a user hovers over it.
`.clear`	`background: #f66;`	Sets the background color for the "Clear" button to a shade of red.
`.operator`	`background: #6c0;`	Sets the background color for operator buttons (`+`, `-`, etc.) to a shade of green.
	`color: #fff;`	Sets the text color to white for operator buttons.

`.equals`	`background: #66f;`	Sets the background color for the "Equals" button to a shade of blue.
	`grid-row: span 2;`	Makes the "Equals" button span across two rows in the grid layout.

⇒ JavaScript Code Details

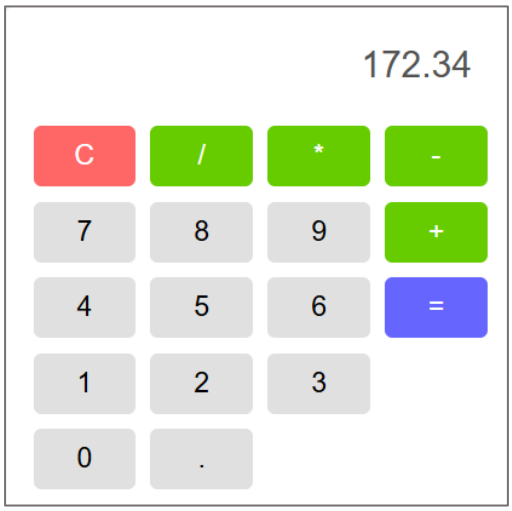
Element/Concept	Description/Purpose
`display` **variable (Line 1)**	Stores a reference to the calculator's screen element in the HTML Document Object Model (DOM).
`appendToDisplay()` **function (L 3-5)**	Appends a number or operator `value` to the current content of the display field.
`clearDisplay()` **function (L 6-8)**	Resets the display field's value to an empty string `''` (clears the screen).
`calculate()` **function (L 10-17)**	Calculates the final result of the expression currently in the display field.
`try...catch` **block (L 11-16)**	A control structure used to run code safely; if an error occurs (e.g., invalid math), it "catches" the error.
`eval()` **(Line 13)**	A JavaScript function that reads a string argument as code and executes it, returning the result of the calculation.
`'Error'` **(Line 15)**	The message displayed on the calculator screen if the `eval()` function encounters an invalid mathematical expression.

CODE OUTPUT:



- Enter random numbers for **division**. Press **C** (clear) for next **operation**.

- Enter random numbers for **Addition**.



- Enter random numbers for **Multiplication**.